

PERMUTATIONS

In the last lesson, you counted the possible ways that four children could line up for a photograph. A way was considered different if the children were in a different order. When order matters, you are counting the number of **permutations**.

In real life, the word **combination** is often used when really it is a **permutation**. When you use a combination lock, you have to enter the numbers in a certain order. A combination lock should really be called a permutation lock.

You will learn more about combinations in the next lesson. For now, remember that in a permutation problem, the order of the events or items matters.

Permutation lock

10 - 15 - 30



COUNTING PERMUTATIONS

Example In how many different ways can a student arrange five different books on a shelf?

Step 1

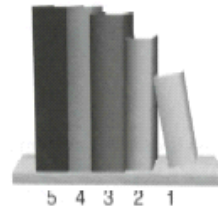
Count the number of choices the student has for each space on the shelf.

There are 5 choices for the first book. After a book is placed in that position, there are 4 choices for the second book. After each book is placed, there is 1 fewer choice. Notice how the choices count down after each book is placed.

Step 2

Multiply the choices: $5 \times 4 \times 3 \times 2 \times 1 = 120$

There are 120 different ways the books can be arranged on the shelf.



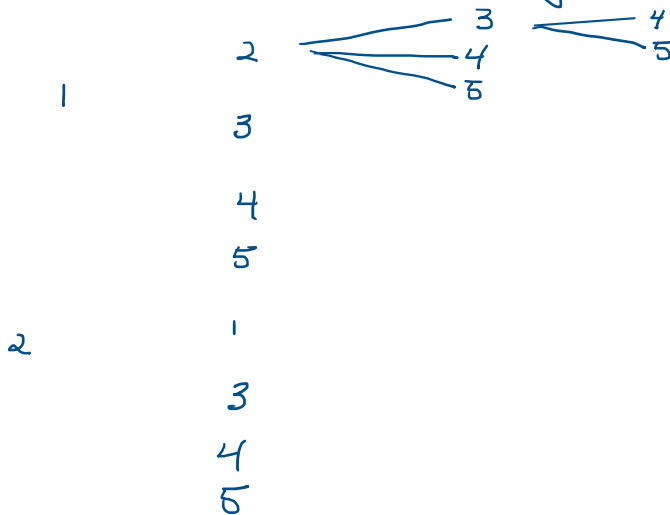
Recall the 4 children in the family photo?

$4 \times 3 \times 2 \times 1 = 24$ ways

OR

$$P(4,4) = \frac{4!}{0!} = 24$$

If we created a probability tree, it would look like this:



$$\begin{aligned} \text{OR} \\ P(5,5) &= \frac{5!}{0!} \\ &= 120 \end{aligned}$$

Permutations, cont'd

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To solve any permutation problem, simply multiply the choices. Study how this basic idea can be used to solve a more complex problem.

FINDING PERMUTATIONS OF PART OF A GROUP

Example Fifteen runners are in a race. How many different finishes can there be for 1st, 2nd, and 3rd place?

Step 1

Count the number of choices. There are 15 runners who could finish in 1st place, 14 choices for 2nd place, and 13 choices for 3rd place. Remember, once a runner finishes 1st, that runner cannot also finish in 2nd or 3rd place.



Step 2

Multiply the choices: $15 \times 14 \times 13 = 2,730$ ✓

There are 2,730 different ways the runners could finish 1st, 2nd, and 3rd.

Will the choices always count down? Not always.

Some choices can be reused

Example A bike combination lock has three dials, each with 10 numbers. To open the lock, you turn the dials. A number can be used more than once. How many permutations are possible?

There are 10 choices for each dial: $10 \times 10 \times 10 = 1,000$.



Examples

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Solve.

- ✓ 1. Jasmine has 8 songs in a playlist. The playlist is set to play songs randomly. If she listens to 3 songs, how many different orders are possible if a song may not be repeated?

$$8 \times 7 \times 6 = 336 \text{ ways}$$

$$\text{OR: } P(8, 3) = \frac{8!}{(8-3)!} = \frac{8!}{5!}$$

$$= \frac{8 \times 7 \times 6 \times 5!}{5!} = 336$$

- ✓ 2. A company assigns a code to each customer. The code has 2 letters followed by a single-digit number from 1 to 9. If the letters must be different, how many codes can the company make? (Hint: There are 26 letters in the alphabet.)

$$26 \times 25 \times 9 = 5850$$

$$\text{OR: letters } P(26, 2) = \frac{26!}{24!} = 26 \times 25$$

$$= 650$$

then each of these 650 has 1 number
so, $650 \times 9 = 5850$

- ✓ 3. For summer reading, Grace must choose 4 books out of a reading list of 12 books. In how many different ways could Grace complete the assignment?

<u>1st choice</u>	<u>2nd choice</u>	<u>3rd choice</u>	<u>4th choice</u>
12 Books	11 books	10 Books	9 Books

$$12 \times 11 \times 10 \times 9 = 11,880 \text{ ways}$$

$$\text{OR, } P(12, 4) = \frac{12!}{8!} = 12 \times 11 \times 10 \times 9 = 11880$$

- ✓ 4. Five friends are performing a song for a talent show. They plan to stand in a line. In how many different orders could they stand?

$$5 \times 4 \times 3 \times 2 \times 1 = 120 \text{ ways}$$

$$\text{OR } P(5, 5) = \frac{5!}{0!} = 120$$